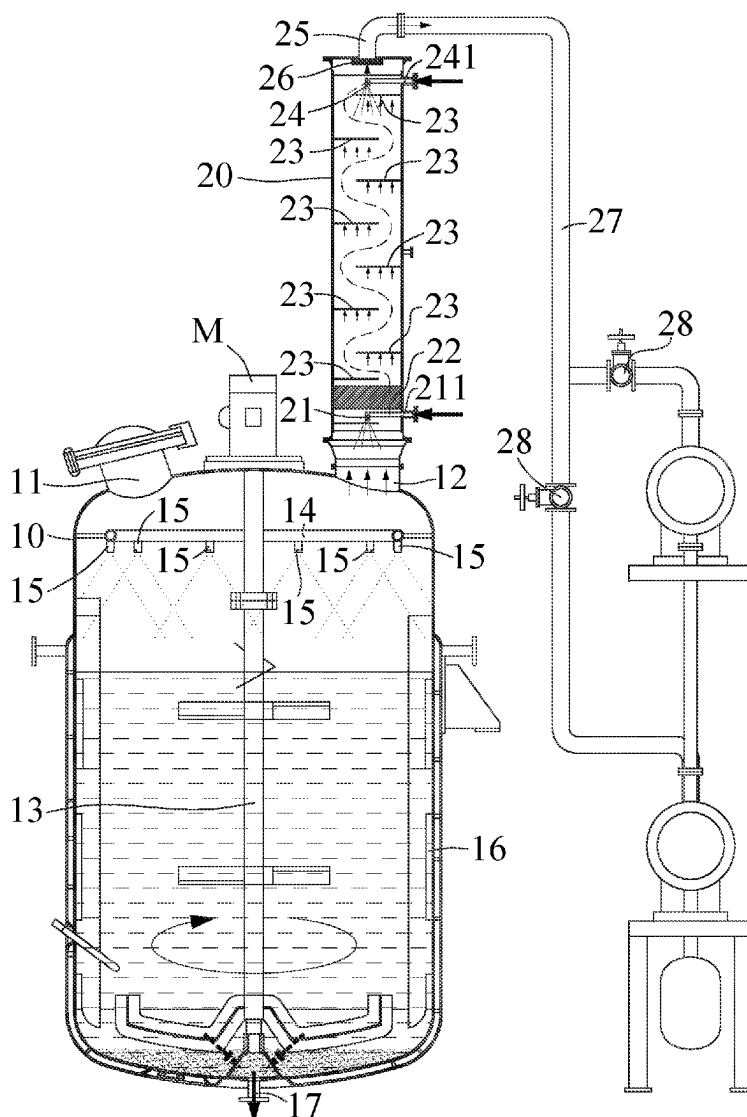




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(19) **United States**(12) **Patent Application Publication**
Liu(10) **Pub. No.: US 2025/0269297 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **RECYCLING AND PURIFYING STRUCTURE
FOR DISTILLER**(57) **ABSTRACT**(71) Applicant: **Feng-Yu Liu**, Taichung City (TW)(72) Inventor: **Feng-Yu Liu**, Taichung City (TW)(21) Appl. No.: **18/583,931**(22) Filed: **Feb. 22, 2024****Publication Classification**(51) **Int. Cl.****B01D 3/14** (2006.01)**B01D 5/00** (2006.01)(52) **U.S. Cl.**CPC **B01D 3/14** (2013.01); **B01D 5/006**
(2013.01)

A recycling and purifying structure contains: a storage bucket configured to store waste liquids and a distilling body configured to distill the waste liquids. The storage bucket includes an inlet defined and an outlet. The distilling body is connected with the outlet and includes a first atomizing nozzle which has a feeding opening configured to produce large-molecular recycled waters, a second atomizing nozzle having a returning mouth for producing small-molecular recycled waters from the large-molecular recycled waters of the feeding opening. The distilling body further includes multiple filtering elements separately disposed between the first atomizing nozzle and the second atomizing nozzle, and a respective one filtering element has multiple screen elements. The distilling body includes a recycling tube connected on a center of a top of an outer wall thereof, and a bubble absorber mounted on the recycling tube to eliminate air bubbles of the small-molecular recycled waters.



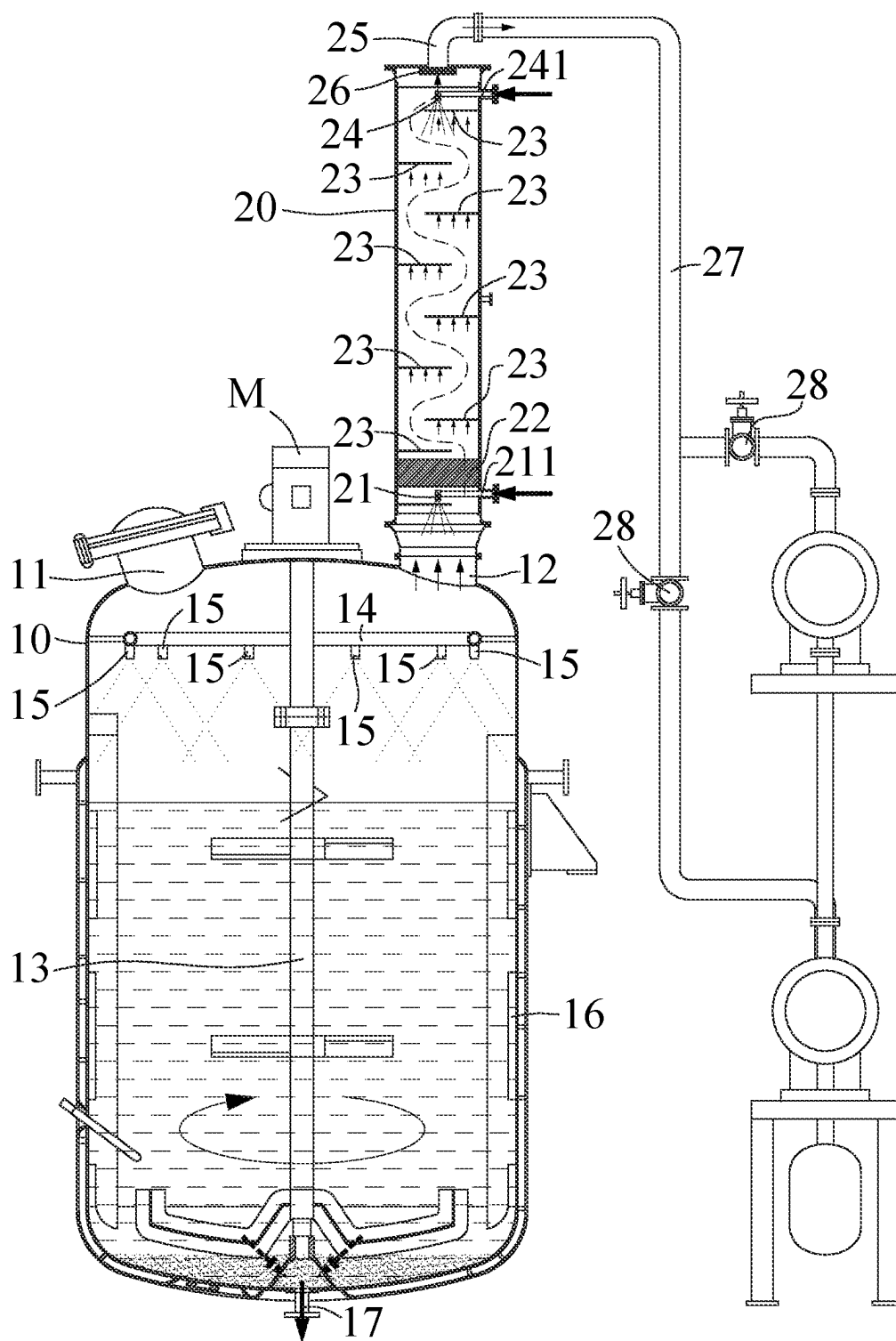


FIG 1

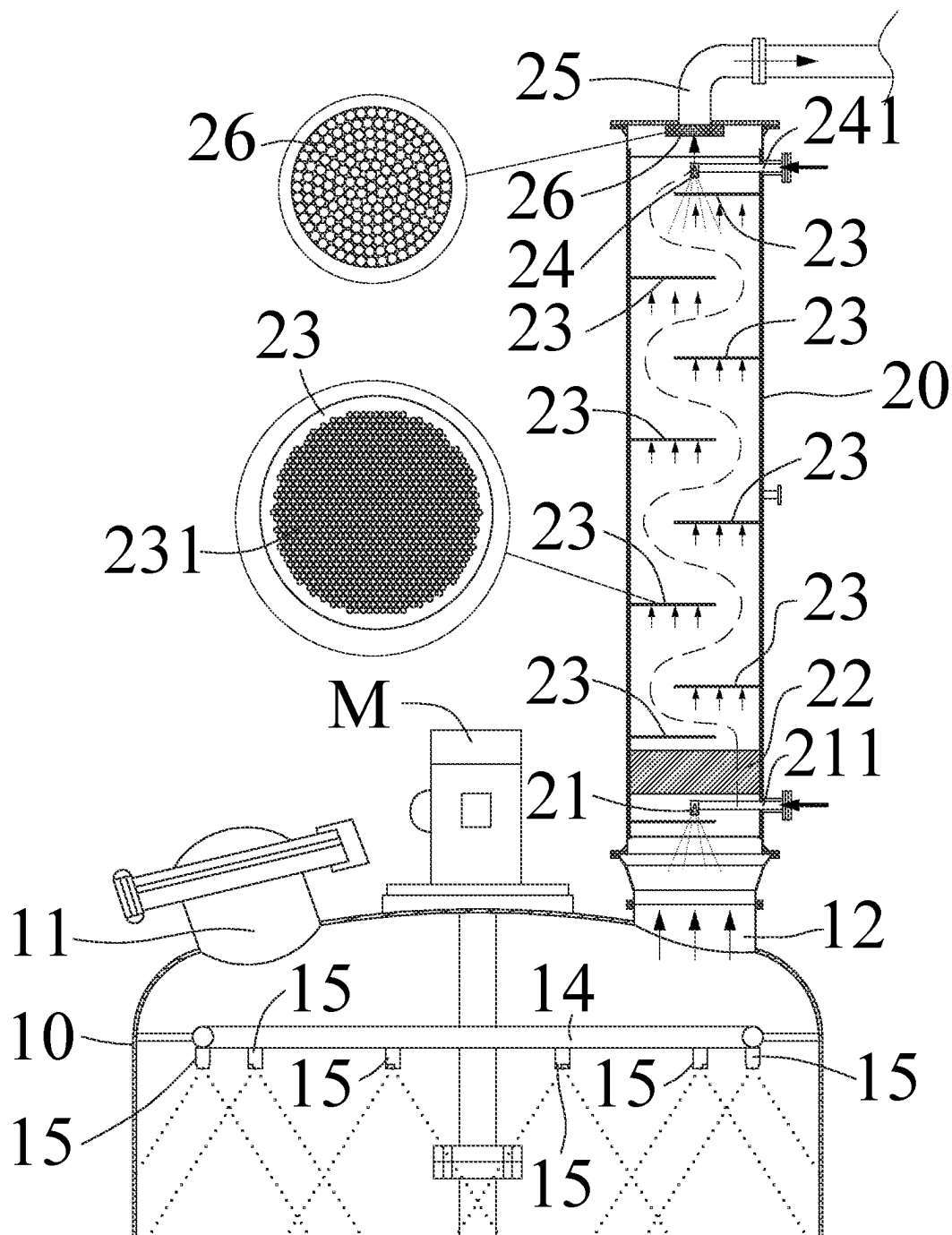


FIG. 2

RECYCLING AND PURIFYING STRUCTURE FOR DISTILLER

TECHNICAL FIELD

[0001] The present disclosure relates to a recycling and purifying structure for a distiller which is capable of accelerating recycling and purifying waste liquids.

BACKGROUND

[0002] Taiwan is a major semiconductor manufacturing country and contributes considerable output value every year, but it also produces a large amount of waste. Wastewater produced by IC manufacturing plants, and most of wastewater are discharged from cleaning wafers, removing photoresist and etching. In the era of circular economy, how to convert waste into raw materials is something that both industry and government should pay attention to.

[0003] A conventional distilling method and a distiller are disclosed in TW Patent No. 157517 and are applied to decrease a boiling point of waste waters in the stiller within 30° C. to 60° C., wherein the waste waters are vacuumed in the stiller and are condensed to waters from steams by a condenser.

[0004] However, such a method and distiller cannot produce recycled waters in small molecules from large molecules of the waste waters. In addition, the recycled waters cannot have high purity and flows in a quick flow rate.

[0005] The present invention has arisen to mitigate and/or obviate the afore-described disadvantages.

SUMMARY

[0006] A primary aspect of the present invention is to provide a recycling and purifying structure for a distiller which is capable of accelerating recycling and purifying waste liquids.

[0007] To obtain above-mentioned aspect, a recycling and purifying structure provided by the present invention contains: a storage bucket configured to store waste liquids and a distilling body configured to distill the waste liquids.

[0008] The storage bucket includes an inlet defined on a top of an outer wall thereof, and an outlet formed on the top of the outer wall of the storage bucket opposite to the inlet.

[0009] The distilling body is connected with the outlet of the storage bucket, and the distilling body includes a first atomizing nozzle which has a feeding opening configured to produce large-molecular recycled waters, a second atomizing nozzle arranged on a side of a top thereof and having a returning mouth for producing small-molecular recycled waters from the large-molecular recycled waters of the feeding opening.

[0010] The distilling body further includes multiple filtering elements separately disposed between the first atomizing nozzle and the second atomizing nozzle, and a respective one filtering element has multiple screen elements. The distilling body includes a recycling tube connected on a center of a top of an outer wall thereof, and a bubble absorber mounted on the recycling tube to eliminate air bubbles of the small-molecular recycled waters.

[0011] Preferably, the storage bucket further includes a motor fixed on a center of the top of the outer wall thereof and configured to drive a stirrer to rotate. The stirrer is vertically inserted into a bottom of an inner wall of the storage bucket and is configured to rotatably stir the waste

liquids evenly, the storage bucket includes a circular tube fixed on a top of the inner wall thereof, multiple spray heads defined on the circular tube and configured to molecularize the waste liquids, multiple heaters arranged on the outer wall of the storage bucket away from the motor and configured to heat the water liquids in the storage bucket, and a discharge orifice defined on a center of a bottom of the inner wall of the storage bucket and configured to discharge the waste liquids.

[0012] Preferably, the distilling body further includes a filtration layer arranged above the first atomizing nozzle.

[0013] Preferably, the respective one filtering element is circular.

[0014] Preferably, the multiple screen elements are through holes.

[0015] Preferably, the multiple screen elements are meshes.

[0016] Other objects and features will be in part apparent and in part pointed out hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a cross sectional view showing the assembly of a recycling and purifying structure for a distiller according to a preferred embodiment of the present invention.

[0018] FIG. 2 is a schematic view showing the operation of the recycling and purifying structure for the distiller according to the preferred embodiment of the present invention.

DETAILED DESCRIPTION

[0019] FIG. 1 is a cross sectional view showing the assembly of a recycling and purifying structure for a distiller according to a preferred embodiment of the present invention. The recycling and purifying structure comprises: a storage bucket 10 configured to store waste liquids and a distilling body 20 configured to distill the waste liquids.

[0020] The storage bucket 10 includes an inlet 11 defined on a top of an outer wall thereof, an outlet 12 formed on the top of the outer wall of the storage bucket 10 opposite to the inlet 11, a motor M fixed on a center of the top of the outer wall of the storage bucket 10 and configured to drive a stirrer 13 to rotate, wherein the stirrer 13 is vertically inserted into a bottom of an inner wall of the storage bucket 10 and is configured to rotatably stir the waste liquids evenly, the storage bucket 10 also includes a circular tube 14 fixed on a top of the inner wall thereof, multiple spray heads 15 defined on the circular tube 14 and configured to molecularize the waste liquids, multiple heaters 16 arranged on the outer wall of the storage bucket 10 away from the motor M and configured to heat the water liquids in the storage bucket 10, and a discharge orifice 17 defined on a center of a bottom of the inner wall of the storage bucket 10 and configured to discharge toxic solvents or cleaning fluids of the waste liquids, wherein the distilling body 20 is connected with the outlet 12 of the storage bucket 10.

[0021] The distilling body 20 includes a first atomizing nozzle 21 which has a feeding opening 211 configured to produce large-molecular recycled waters, a filtration layer 22 arranged above the first atomizing nozzle 21, and multiple filtering elements 23 separately disposed above the filtration layer 22, wherein a respective one filtering element 23 is circular and has multiple screen elements 231 which

are through holes or honeycomb-shaped meshes so that recycled waters flow upward in an S-shaped path, thus molecularizing the recycled waters from the waste liquids. The distilling body **20** further includes a second atomizing nozzle **24** arranged on a side of a top thereof and having a returning mouth **241** for producing small-molecular recycled waters from the large-molecular recycled waters of the feeding opening **211**, thus accelerating a flow rate of the recycled waters. The distilling body **20** includes a recycling tube **25** connected on a center of a top of an outer wall thereof, and a bubble absorber **26** mounted on the recycling tube **25** to eliminate air bubbles of the recycled waters, wherein the recycling tube **25** is connected with the recycling and purifying equipment via at least one connection pipe **27** and at least valve **28** to recycle the recycled waters from the waste liquids.

[0022] FIG. 2 is a schematic view showing the operation of the recycling and purifying structure for the distiller according to the preferred embodiment of the present invention. The toxic solvents or cleaning fluids of the waste liquids in the storage bucket **10** are heated to produce steams by the stilling body **20**. When the steams float upward, the feeding opening **211** of the first atomizing nozzle **21** sprays the large-molecular recycled waters from the recycled waters, thus enhancing purity of the recycled waters. In the meantime, the steams are sprayed out of the second atomizing nozzle **24** to produce the small-molecular recycled waters after flowing through the filtration layer **22**, the S-shaped path, the multiple filtering elements **23** and the multiple screen elements **231**, thus increasing the flow rate of the recycled waters. Thereafter, the small-molecular recycled waters produces the air bubbles and the air bubbles are eliminated by the bubble absorber **26** so as to be recycled via the at least one connection pipe **27** and the at least valve **28** from the recycling tube **25**.

[0023] When introducing elements of the present invention or the preferred embodiments thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

[0024] In view of the above, it will be seen that the several objects of the invention are achieved and other advantageous results attained.

[0025] As various changes could be made in the above constructions, products, and methods without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

What is claimed is:

1. A recycling and purifying structure comprising: a storage bucket configured to store waste liquids and a distilling body configured to distill the waste liquids;

wherein the storage bucket includes an inlet defined on a top of an outer wall thereof, and an outlet formed on the top of the outer wall of the storage bucket opposite to the inlet;

wherein the distilling body is connected with the outlet of the storage bucket, and the distilling body includes a first atomizing nozzle which has a feeding opening configured to produce large-molecular recycled waters, a second atomizing nozzle arranged on a side of a top thereof and having a returning mouth for producing small-molecular recycled waters from the large-molecular recycled waters of the feeding opening;

wherein the distilling body further includes multiple filtering elements separately disposed between the first atomizing nozzle and the second atomizing nozzle, and a respective one filtering element has multiple screen elements, wherein the distilling body includes a recycling tube connected on a center of a top of an outer wall thereof, and a bubble absorber mounted on the recycling tube to eliminate air bubbles of the small-molecular recycled waters.

2. The recycling and purifying structure as claimed in claim **1**, wherein the storage bucket further includes a motor fixed on a center of the top of the outer wall thereof and configured to drive a stirrer to rotate, wherein the stirrer is vertically inserted into a bottom of an inner wall of the storage bucket and is configured to rotatably stir the waste liquids evenly, the storage bucket includes a circular tube fixed on a top of the inner wall thereof, multiple spray heads defined on the circular tube and configured to molecularize the waste liquids, multiple heaters arranged on the outer wall of the storage bucket away from the motor and configured to heat the water liquids in the storage bucket, and a discharge orifice defined on a center of a bottom of the inner wall of the storage bucket and configured to discharge the waste liquids.

3. The recycling and purifying structure as claimed in claim **1**, wherein the distilling body further includes a filtration layer arranged above the first atomizing nozzle.

4. The recycling and purifying structure as claimed in claim **1**, wherein the respective one filtering element is circular.

5. The recycling and purifying structure as claimed in claim **1**, wherein the multiple screen elements are through holes.

6. The recycling and purifying structure as claimed in claim **1**, wherein the multiple screen elements are meshes.

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